

Problem 1

1. C 2. B 3. A 4. B 5. A

Problem 3

Compute Action: $A|0\rangle = |0\rangle$, $A|1\rangle = -|1\rangle$, $B|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$, $B|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$

$$\therefore A \otimes B \cdot |01\rangle = |0\rangle \otimes \frac{|0\rangle - |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}} (|00\rangle - |01\rangle)$$

$$A \otimes B \cdot |10\rangle = -|1\rangle \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} = -\frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

$$\Rightarrow A \otimes B \cdot |\psi\rangle = \frac{1}{\sqrt{2}} \left(\frac{|00\rangle - |01\rangle}{\sqrt{2}} + \frac{|10\rangle + |11\rangle}{\sqrt{2}} \right) = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle + |11\rangle)$$

$$E(A, B) = \frac{1}{2} (\langle 01| - \langle 10|) \cdot \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle + |11\rangle) = \frac{1}{2\sqrt{2}} (-1+1) = 0$$

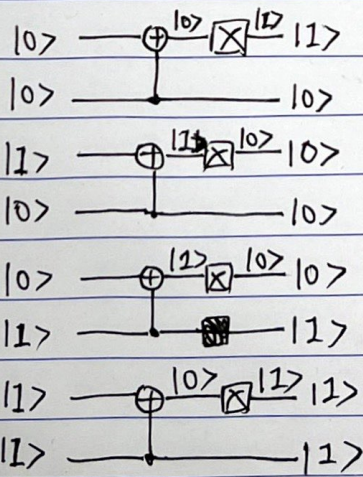
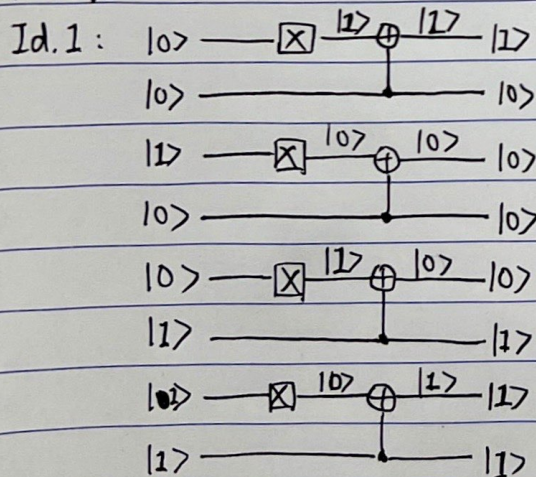
$$\text{Else can get: } E(A_2, B_1) = \frac{1}{2} (\langle 01| - \langle 10|) \cdot \frac{1}{2} (-|00\rangle - |01\rangle + |10\rangle - |11\rangle) = \frac{1}{2\sqrt{2}} (-1+1) = 0$$

$$E(A_1, B_2) = \frac{1}{2} (\langle 01| - \langle 10|) \cdot \frac{1}{2} (-|00\rangle + |01\rangle + |10\rangle - |11\rangle) = \frac{1}{2\sqrt{2}} (1+1) = \frac{1}{\sqrt{2}}$$

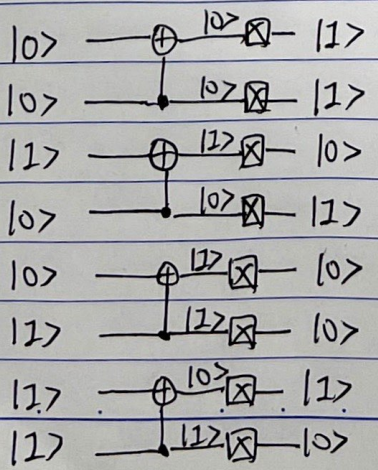
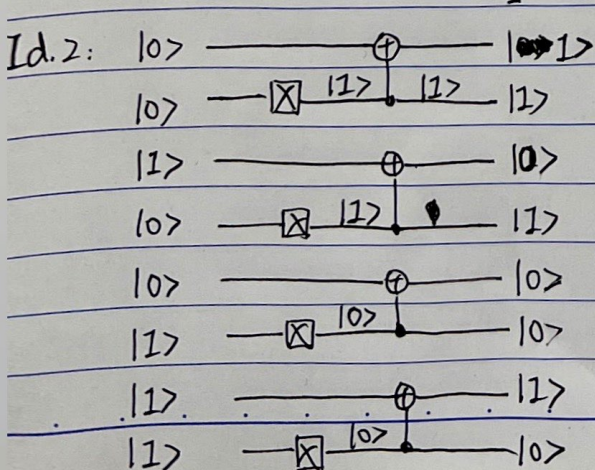
$$E(A_2, B_2) = \frac{1}{2} (\langle 01| - \langle 10|) \cdot \frac{1}{2} (-|00\rangle + |01\rangle - |10\rangle - |11\rangle) = \frac{1}{2\sqrt{2}} (-1-1) = -\frac{1}{\sqrt{2}}$$

$$\therefore S = 0 + 0 + \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$$

Problem 4



All inputs lead to the same outputs, proved.



All inputs lead to the same outputs, proved